UIT 1711 MOBILE APPLICATION DEVELOPMENT LAB

A LABORATORY RECORD
(MODULE 1)

Submitted by

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VII SEMESTER B.TECH IT

DEPARTMENT OF INFORMATION TECHNOLOGY

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING
(AFFILIATED TO ANNA UNIVERSITY CHENNAI)

NOV 2021

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LAB RECORD

UIT1711-Mobile Application Development Lab

Date: 13-09-2021

Certified that this is the bonafide record work done by

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Academic Year	2021- 2022
Laboratory	UIT1711-MOBILE APPLICATION DEVELOPMENT LAB

SIGNATURE OF THE STUDENT **FACULTY IN CHARGE**

Hello World

EX.No: 1

Aim:

To create a Hello world application using flutter

Code:

```
import 'dart:ui';

import 'package:flutter/material.dart';

void main() {
runApp(new MaterialApp(
   home: new MyApp(),
  ));
}

class MyApp extends StatefulWidget {
  @override
  _State createState() => new _State();
}

class _State extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
   return new Scaffold(
appBar: new AppBar(
       title: new Text('My App'),
     ),
     body: new Center(
         child: new Column(
           children: <Widget>[
             //default color
             new Text(
               'Hello world',
               style: TextStyle(
fontSize: 35,
```

```
            ),
          ),
          new Text(
            'Hello world',
            style: TextStyle(
              //inbuilt named colors
              color: Colors.blue,
fontSize: 40,
            ),
          ),
          new Text(
            'Hello world',
            style: TextStyle(
              //using ARGB (Alpha Red Green Blue)
              color: Color.fromARGB(255, 255, 0, 0),
fontSize: 55,
            ),
          ),
          new Text(
            'Hello world',
            style: TextStyle(
              //using RGBO (Red Green Blue Opacity)
              color: Color.fromRGBO(0, 155, 0, 0.8),
fontSize: 75
            ),
          ),
        ],
      )
    ),
  );
 }
}
```

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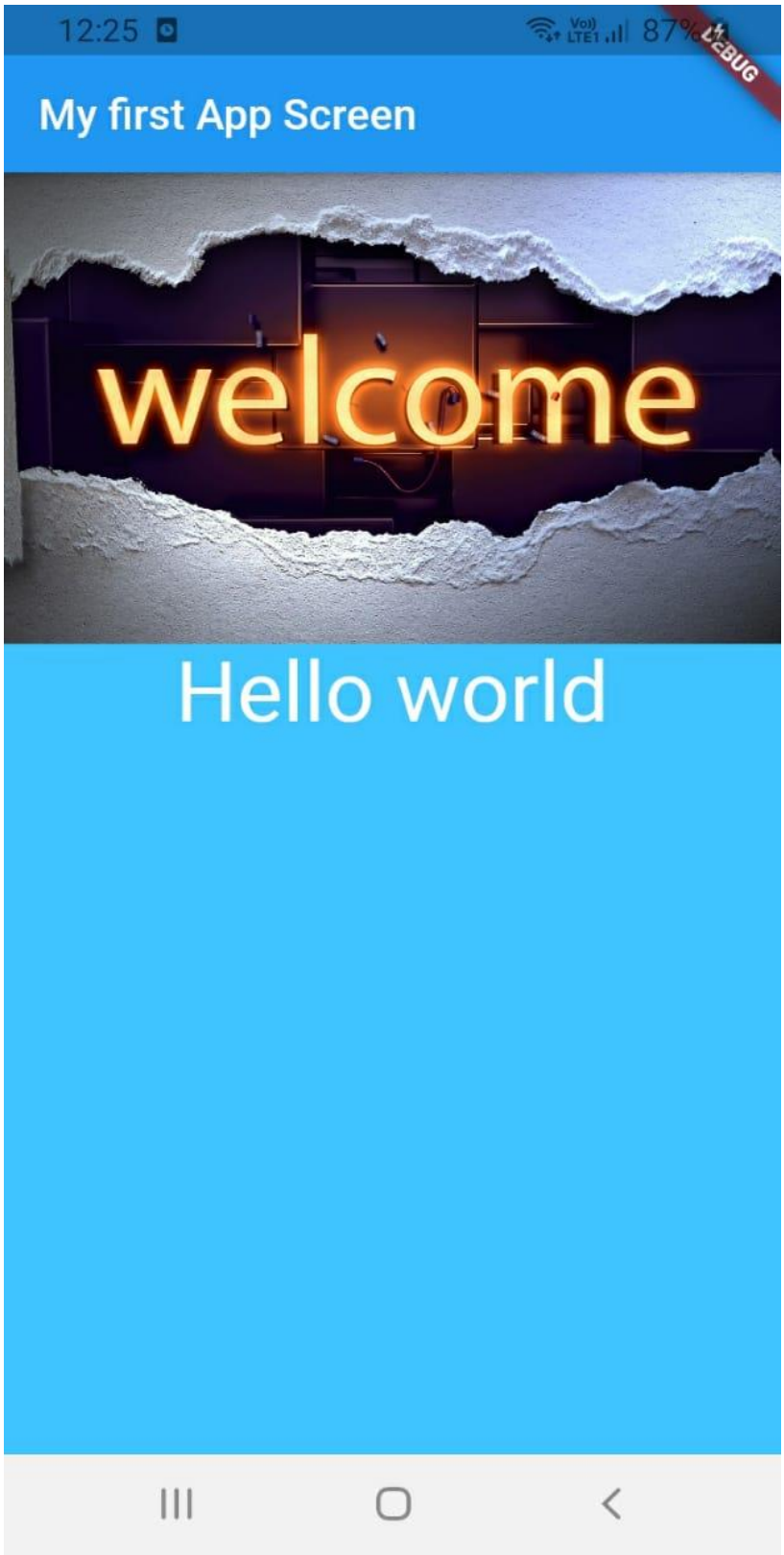
Output:



Result:

The hello world application is created successfully.

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Image

Ex.No:2

Aim:

To create a image application using flutter

Code:

```
import 'package:flutter/material.dart';

class Images extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
backgroundColor: Colors.amber,
appBar: AppBar(
        title: Row(
          children: [
CircleAvatar(
backgroundImage: AssetImage('assets/images/hibiscus.jpeg'),
            ),
SizedBox(
              width: 15,
            ),
            Text('Flowers'),
          ],
        ),
centerTitle: true,
backgroundColor: Colors.black,
      ),
      body: Center(
        child: Column(
          children: [
Container(
              width: 1000,
              height: 50,
              margin: const EdgeInsets.only(bottom: 40),
              color: Colors.redAccent,
              child: Center(
                child: Text(
                  'Harry Potter',
                  style: TextStyle(fontStyle: FontStyle.italic, fontSize: 25),
                ),
```

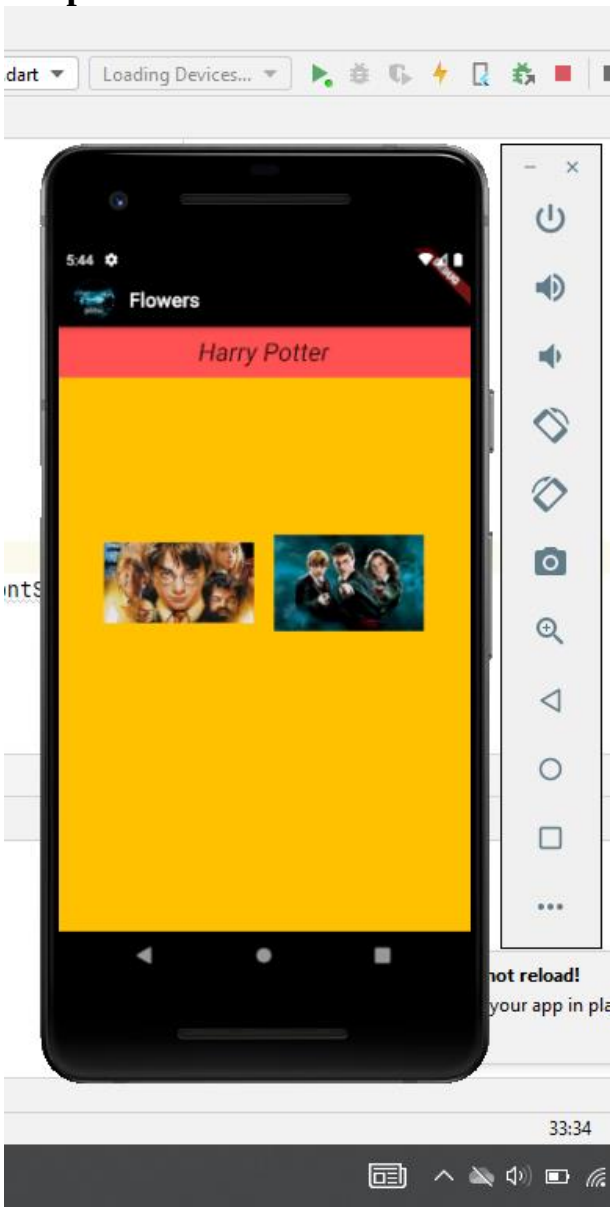
```
            ),
          ),
Padding(
            padding: const EdgeInsets.all(15.0),
            child: Row(
mainAxisAlignment: MainAxisAlignment.center,
              children: [
ClipRRect(
borderRadius: BorderRadius.circular(100),
                  child: Image.asset(
                    'assets/images/flower.jpeg',
                    height: 300,
                    width: 150,
                  )),
SizedBox(
                width: 20,
              ),
ClipRRect(
borderRadius: BorderRadius.circular(100),
                child: Image.asset(
                  'assets/images/decor.jpeg',
                  height: 300,
                  width: 150,
                ),
              ),
            ],
          ),
        ),
      ],
    ),
  ),
  );
 }
}


import 'package:flutter/material.dart';
import 'package:image/ui/images.dart';

void main() {
runApp(MaterialApp(
   title: 'Images',
```

```
    home: Images(),
  ));
}
```

Output:



Result:

The image application is created successfully.

Draw using flutter

Ex.No:3

Aim:

To draw shapes using flutter

Code:

```
import 'package:flutter/material.dart';
import 'package:sample/ui/traffic.dart';

void main() {
runApp(new MaterialApp(
   title: "SSN Workshop",
   home: Traffic(), // pass the name of the class, which will be the initial class
flutter will load
  ));
}


import 'package:flutter/material.dart';
class Traffic extends StatelessWidget {
  const Traffic({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
   return Scaffold(
appBar: AppBar(
       title: Text('Traffic Lights'),
backgroundColor: Colors.lightGreen,
     ),
body:Center(
       child: Container(
         width: 100,
         height: 220,
         color: Colors.black,
         child: Column(
mainAxisAlignment: MainAxisAlignment.center,
crossAxisAlignment: CrossAxisAlignment.center,
           children: [
Container(
               width: 50,
               height: 50,
               decoration: BoxDecoration(
                 shape: BoxShape.circle, color: Colors.red
```

```
            ),
          ),
SizedBox(
            height:15
          ),
Container(
            width: 50,
            height: 50,
            decoration: BoxDecoration(
              shape: BoxShape.circle, color: Colors.amber
            ),
          ),
SizedBox(
              height:15
          ),
Container(
            width: 50,
            height: 50,
            decoration: BoxDecoration(
              shape: BoxShape.circle,color: Colors.green
            ),
          ),
        ],
      ),
    ),
  )
 );
 }
}
```

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Output:

Result:

The shapes are drawn using flutter success

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Form Creation and Validation

Ex.No:4

Aim:

To create a form creation and validation using flutter

Code:

Creation:

```
import 'package:flutter/material.dart';

void main() =>runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final appTitle = 'Flutter Form Demo';
    return MaterialApp(
      title: appTitle,
      home: Scaffold(
appBar: AppBar(
          title: Text(appTitle),
        ),
        body: MyCustomForm(),
      ),
    );
  }
}
// Create a Form widget.
class MyCustomForm extends StatefulWidget {
  @override
MyCustomFormStatecreateState() {
    return MyCustomFormState();
  }
}
// Create a corresponding State class. This class holds data related to the form.
class MyCustomFormState extends State<MyCustomForm> {
  // Create a global key that uniquely identifies the Form widget
  // and allows validation of the form.
  final _formKey = GlobalKey<FormState>();

  @override
```

```
  Widget build(BuildContext context) {
    // Build a Form widget using the _formKey created above.
    return Form(
      key: _formKey,
      child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
        children: <Widget>[
TextFormField(
            decoration: const InputDecoration(
              icon: const Icon(Icons.person),
hintText: 'Enter your name',
labelText: 'Name',
            ),
          ),
TextFormField(
            decoration: const InputDecoration(
              icon: const Icon(Icons.phone),
hintText: 'Enter a phone number',
labelText: 'Phone',
            ),
          ),
TextFormField(
            decoration: const InputDecoration(
            icon: const Icon(Icons.calendar_today),
hintText: 'Enter your date of birth',
labelText: 'Dob',
            ),
           ),
          new Container(
              padding: const EdgeInsets.only(left: 150.0, top: 40.0),
              child: new RaisedButton(
                child: const Text('Submit'),
onPressed: null,
              )),
        ],
      ),
    );
  }
}
```

Output:

Validation:

```
validator: (value) {
      if (value.isEmpty) {
            return 'Please enter some text';
      }
      return null;
},
```

```
import 'package:flutter/material.dart';

void main() =>runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final appTitle = 'Flutter Form Demo';
    return MaterialApp(
```

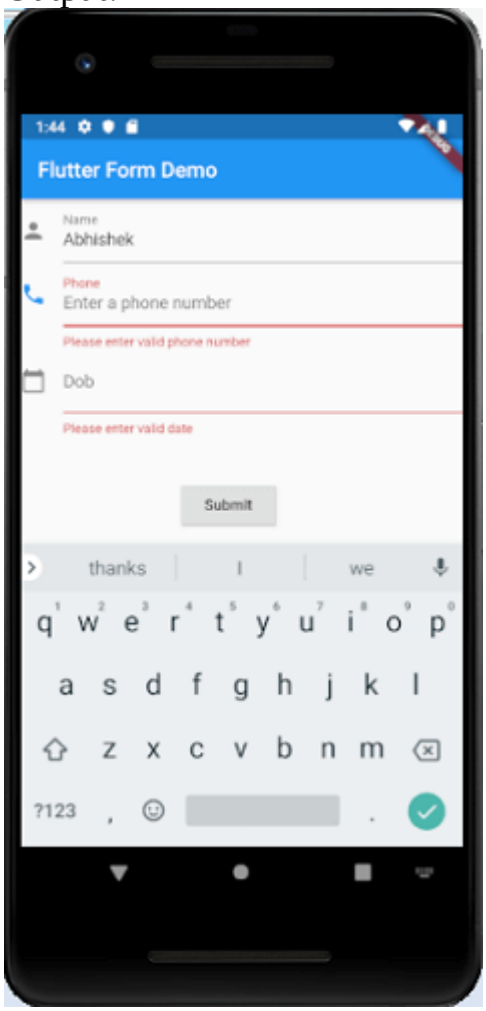
```
      title: appTitle,
      home: Scaffold(
appBar: AppBar(
          title: Text(appTitle),
        ),
        body: MyCustomForm(),
      ),
    );
  }
}
// Create a Form widget.
class MyCustomForm extends StatefulWidget {
  @override
MyCustomFormStatecreateState() {
    return MyCustomFormState();
  }
}
// Create a corresponding State class, which holds data related to the form.
class MyCustomFormState extends State<MyCustomForm> {
  // Create a global key that uniquely identifies the Form widget
  // and allows validation of the form.
  final _formKey = GlobalKey<FormState>();

  @override
  Widget build(BuildContext context) {
    // Build a Form widget using the _formKey created above.
    return Form(
      key: _formKey,
      child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
        children: <Widget>[
TextFormField(
            decoration: const InputDecoration(
              icon: const Icon(Icons.person),
hintText: 'Enter your full name',
labelText: 'Name',
            ),
            validator: (value) {
              if (value.isEmpty) {
                return 'Please enter some text';
              }
              return null;
```

```
          },
        ),
TextFormField(
          decoration: const InputDecoration(
            icon: const Icon(Icons.phone),
hintText: 'Enter a phone number',
labelText: 'Phone',
          ),
          validator: (value) {
            if (value.isEmpty) {
              return 'Please enter valid phone number';
            }
            return null;
          },
        ),
TextFormField(
          decoration: const InputDecoration(
          icon: const Icon(Icons.calendar_today),
hintText: 'Enter your date of birth',
labelText: 'Dob',
          ),
          validator: (value) {
            if (value.isEmpty) {
              return 'Please enter valid date';
            }
            return null;
          },
         ),
        new Container(
            padding: const EdgeInsets.only(left: 150.0, top: 40.0),
            child: new RaisedButton(
              child: const Text('Submit'),
onPressed: () {
                // It returns true if the form is valid, otherwise returns false
                if (_formKey.currentState.validate()) {
                  // If the form is valid, display a Snackbar.
Scaffold.of(context)
.showSnackBar(SnackBar(content: Text('Data is in processing.')));
                }
              },
            )),
      ],
```

```
      ),
    );
  }
}
```

Output:



Result:

A form is being created and validated successfully.

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Simple Calculator Using Flutter

Ex.No:5

Aim:

To implement a simple calculator using flutter.

Code:

```
import 'package:flutter/material.dart';
import 'buttons.dart';
import 'package:math_expressions/math_expressions.dart';

void main() {
runApp(MyApp());
}

class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
	return MaterialApp(
	debugShowCheckedModeBanner: false,
	home: HomePage(),
	); // MaterialApp
}
}

class HomePage extends StatefulWidget {
@override
_HomePageStatecreateState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
var userInput = '';
var answer = '';

// Array of button
final List<String> buttons = [
	'C',
	'+/-',
	'%',
	'DEL',
	'7',
	'8',
	'9',
```

```
        '/',
        '4',
        '5',
        '6',
        'x',
        '1',
        '2',
        '3',
        '-',
        '0',
        '.',
        '=',
        '+',
];

@override
Widget build(BuildContext context) {
        return Scaffold(
        appBar: new AppBar(
                title: new Text("Calculator"),
        ), //AppBar
        backgroundColor: Colors.white38,
        body: Column(
                children: <Widget>[
                Expanded(
                        child: Container(
                        child: Column(
                                mainAxisAlignment:
MainAxisAlignment.spaceEvenly,
                                children: <Widget>[
                                        Container(
                                        padding: EdgeInsets.all(20),
                                        alignment: Alignment.centerRight,
                                        child: Text(
                                                userInput,
                                                style: TextStyle(fontSize: 18, color:
Colors.white),
                                        ),
                                        ),
                                        Container(
                                        padding: EdgeInsets.all(15),
                                        alignment: Alignment.centerRight,
```

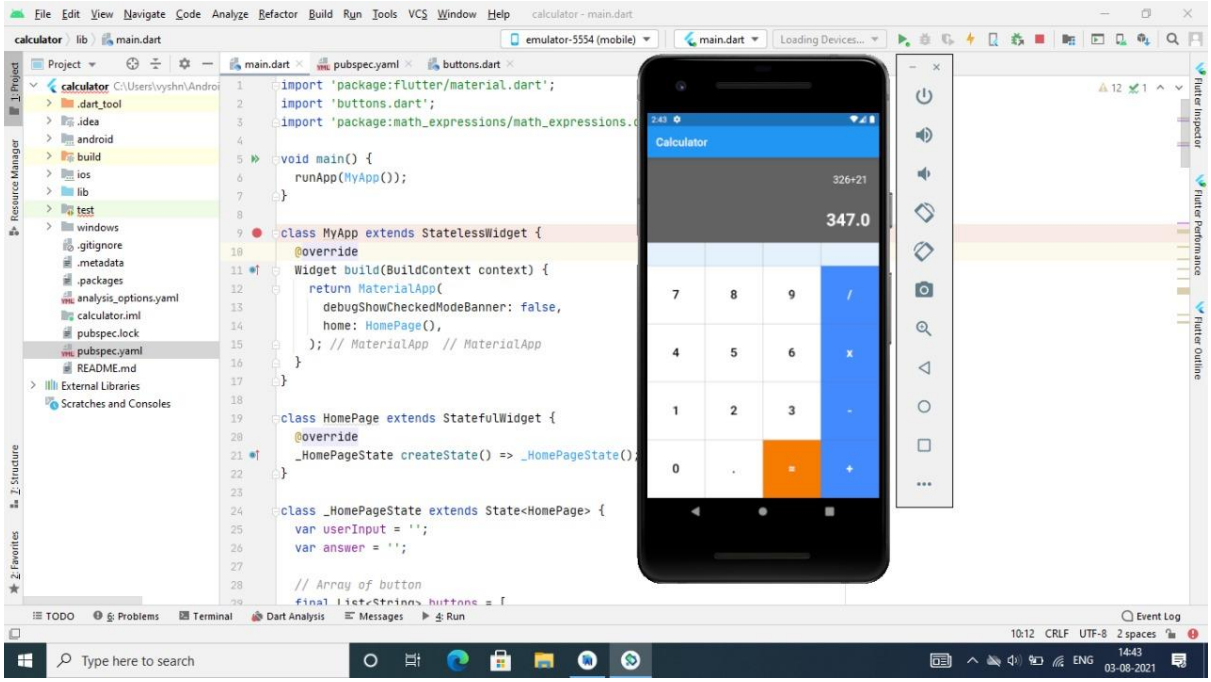
```
                                child: Text(
                                        answer,
                                        style: TextStyle(
                                                fontSize: 30,
                                                color: Colors.white,
                                                fontWeight: FontWeight.bold),
                                ),
                                )
                        ]),
                ),
        ),
        Expanded(
                flex: 3,
                child: Container(
                child: GridView.builder(
                        itemCount: buttons.length,
                        gridDelegate:
SliverGridDelegateWithFixedCrossAxisCount(
                                crossAxisCount: 4),
                        itemBuilder: (BuildContext context, int index) {
                                // Clear Button
                                if (index == 0) {
                                return MyButton(
                                        buttontapped: () {
                                        setState(() {
                                                userInput = '';
                                                answer = '0';
                                        });
                                        },
                                        buttonText: buttons[index],
                                        color: Colors.blue[50],
                                        textColor: Colors.black,
                                );
                                }

                                // +/- button
                                else if (index == 1) {
                                return MyButton(
                                        buttonText: buttons[index],
                                        color: Colors.blue[50],
                                        textColor: Colors.black,
                                );
```

```
}
// % Button
else if (index == 2) {
return MyButton(
	buttontapped: () {
	setState(() {
		userInput += buttons[index];
	});
	},
	buttonText: buttons[index],
	color: Colors.blue[50],
	textColor: Colors.black,
);
}
// Delete Button
else if (index == 3) {
return MyButton(
	buttontapped: () {
	setState(() {
		userInput =
			userInput.substring(0,
userInput.length - 1);
	});
	},
	buttonText: buttons[index],
	color: Colors.blue[50],
	textColor: Colors.black,
);
}
// Equal_to Button
else if (index == 18) {
return MyButton(
	buttontapped: () {
	setState(() {
		equalPressed();
	});
	},
	buttonText: buttons[index],
	color: Colors.orange[700],
	textColor: Colors.white,
);
}
```

```
                                        // other buttons
                                        else {
                                        return MyButton(
                                                buttontapped: () {
                                                setState(() {
                                                        userInput += buttons[index];
                                                });
                                                },
                                                buttonText: buttons[index],
                                                color: isOperator(buttons[index])
                                                        ? Colors.blueAccent
                                                        : Colors.white,
                                                textColor: isOperator(buttons[index])
                                                        ? Colors.white
                                                        : Colors.black,
                                        );
                                        }
                                }), // GridView.builder
                        ),
                ),
                ],
        ),
        );
}

bool isOperator(String x) {
        if (x == '/' || x == 'x' || x == '-' || x == '+' || x == '=') {
        return true;
        }
        return false;
}

// function to calculate the input operation
void equalPressed() {
        String finaluserinput = userInput;
        finaluserinput = userInput.replaceAll('x', '*');

        Parser p = Parser();
        Expression exp = p.parse(finaluserinput);
        ContextModel cm = ContextModel();
        double eval = exp.evaluate(EvaluationType.REAL, cm);
```

```
        answer = eval.toString();
}
}
```

Output:



Result:

Using flutter, a simple calculator is implemented.